

1. ReactJS-HOL

1. Define SPA (Single Page Application)

A **Single Page Application (SPA)** is a web application that loads a single HTML page initially and updates the content dynamically without reloading the entire page.

Benefits of SPA

- Faster page loading after the initial load.
 - Better user experience with smooth navigation.
 - Reduced server requests.
 - Efficient use of APIs.
 - Easy to build responsive web applications.
 - Reusable UI components.
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2. Define React and Identify Its Working

React is an open-source JavaScript library developed by **Meta (Facebook)** for building interactive and dynamic user interfaces, especially Single Page Applications.

Working of React

1. React divides the UI into reusable **components**.
 2. Components are written using **JSX**.
 3. React creates a **Virtual DOM**.
 4. When data changes, React compares the new Virtual DOM with the old one.
 5. Only the changed elements are updated in the Real DOM, making the application faster.
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3. Difference Between SPA and MPA

SPA (Single Page Application) MPA (Multi Page Application)

Loads only one HTML page Loads multiple HTML pages

SPA (Single Page Application) MPA (Multi Page Application)

No full page reload	Reloads every page request
Faster after initial load	Slower due to page reloads
Better user experience	Traditional navigation
Uses AJAX/API for data	Server returns new pages
Examples: Gmail, Facebook	Examples: Amazon, Traditional Banking Websites

4. Pros & Cons of Single Page Application (SPA)

Pros

- Fast performance after initial loading.
- Smooth and responsive user experience.
- Less bandwidth usage.
- Better client-side caching.
- Reusable components.
- Easy integration with REST APIs.

Cons

- Initial loading time can be higher.
 - SEO is more difficult.
 - Heavy dependence on JavaScript.
 - Browser history management is more complex.
 - Debugging may be challenging.
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5. Explain React

React is a JavaScript library used to create modern, interactive, and reusable user interfaces.

Key Characteristics

- Component-based architecture.
- Declarative programming.

- Uses JSX.
- Virtual DOM for faster updates.
- One-way data binding.
- High performance.
- Easy to integrate with other libraries.

Advantages

- Reusable components.
 - Faster rendering.
 - Easy maintenance.
 - Large community support.
 - Excellent performance.
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6. Define Virtual DOM

The **Virtual DOM (Document Object Model)** is a lightweight copy of the Real DOM maintained by React.

How Virtual DOM Works

1. User performs an action.
2. React creates a new Virtual DOM.
3. React compares it with the previous Virtual DOM (Diffing).
4. Only the changed elements are updated in the Real DOM (Reconciliation).

Advantages

- Faster rendering.
 - Improved performance.
 - Reduces direct DOM manipulation.
 - Efficient UI updates.
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7. Features of React

- **Component-Based Architecture** – UI is divided into reusable components.

- **JSX (JavaScript XML)** – Allows writing HTML-like syntax inside JavaScript.
- **Virtual DOM** – Updates only changed parts of the UI.
- **One-Way Data Binding** – Data flows from parent to child components.
- **Declarative UI** – Simplifies UI development.
- **Reusable Components** – Write once and use multiple times.
- **High Performance** – Efficient rendering using Virtual DOM.
- **Easy Learning Curve** – Simple and beginner-friendly.
- **Strong Community Support** – Large ecosystem and extensive documentation.
- **Cross-Platform Development** – Can be used with React Native for mobile apps.